

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python (NumPy, Pandas, Scikit-learn), C++, SQL, Bash

Tools/Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Hugging Face Transformers, Docker, FastAPI, Kubernetes, AWS SageMaker, Git, MLflow, Airflow, Flask

Competence: Machine Learning, Deep Learning, Computer Vision, MLOps, Natural Language Processing, Time Series Prediction, Data Engineering, Perception, Segmentation, Prediction, Planning, Embedded AI, Agile Development

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated **deep learning models** for metrological applications, improving model performance and training efficiency.
- Explored **novel architectures** and **physics-informed hybrid loss functions** to enhance model accuracy and stability, contributing to **state-of-the-art ML techniques**.
- Collaborated with PhD researchers for a **publication at FLOMEKO conference**, demonstrating **scientific background** and contribution.
- Developed interactive dashboards for **data visualization and model performance monitoring**, showcasing **systematic working style** and **data science methods**.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Engineered and deployed **AI algorithms** to achieve 94% **segmentation accuracy** for vitiligo detection, increasing inference speed by 5x; deployed with **Docker** and **FastAPI**, showcasing **perception** capabilities.
- Developed a **real-time tennis foul detection system** leveraging **TensorRT**, reducing compute cost and inference time by 6x, demonstrating expertise in **embedded AI** and **edge processing**.
- Managed **MLOps pipelines** on **AWS SageMaker**, addressing **YOLO model data drift** and maintaining **Kubernetes infrastructure**, ensuring robust model performance and **systematic, goal-oriented development**.
- Developed **proof-of-concept (POC) solutions** using **YOLO**, **few-shot learning**, and **CNNs** for client demonstrations in **perception** and other related applications.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed and deployed an **AI chatbot application** using the **OpenAI API** and modern **LLM tools/frameworks**, demonstrating proficiency in **modern ML techniques** like **Transformers**.
- Engineered a full-stack website providing **AI-based services** using **Flask**, **HTML**, and **CSS**, showcasing strong **Python** development and software engineering skills.
- Implemented solutions requiring **systematic problem-solving** and rapid learning in a dynamic startup environment.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Built and optimized **web scrapers and ETL pipelines** for structured and unstructured data, demonstrating expertise in **handling datasets** and **data science methods**.
- Automated **data ingestion and cleaning workflows** for client analytics, ensuring data quality and supporting **learning-based decision making**.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a **weighted-probability ensemble** integrating DPFunc and DeepGOPlus, utilizing AlphaFold for **feature extraction** and taxonomic bias mitigation, demonstrating **deep neural network** expertise.
- Increased model accuracy by 4% through **custom feature engineering** across MF, CC, and BP namespaces, showcasing **analytical skills** and **state-of-the-art ML techniques**.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed a novel **real-time Bitcoin price forecasting system** using **modern ML techniques**, including recurrent neural networks (RNNs), for **time series prediction**.
- Implemented robust data processing and model evaluation pipelines, ensuring high-frequency predictions and demonstrating **systematic, outcome-oriented development**.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Collaborated to build the world's first **AI-driven self-playing Chitralli Sitar**, combining **signal processing** and **music generation LLMs (Transformers)**, applicable to **time series prediction**.
- Project featured on Coke Studio, showcasing innovative application of **modern ML techniques** and ability to transfer knowledge to new tasks.

Hugging Face

Open-Source

Contributor

- Co-authored the **Zero-shot Image Classification technical documentation** and implementation page on the Hugging Face Hub, demonstrating expertise in **DNNs** and **perception**.
- Contributed to a widely used **AI development framework**, showcasing **scientific background** and open-source collaboration.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialization in Deep Learning, Computer Vision, and Reinforcement Learning, aligning with advanced **ML techniques** for **perception** and **planning**.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor for academic excellence, demonstrating **excellent analytical skills** and a strong **mathematical background**.
- Focused coursework in algorithms, data structures, and foundational computer science principles.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **TensorFlow:** Developer Certificate |

Databricks: Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python